Powers Puzzle Race

Unit 6: Music Studio · Lesson 6-1 · Flagship · EduWonderLab

UNIT 6 · LESSON 6-1 ★ FLAGSHIP · TEACHER NOTES | **Standard 6.EE.1**

FORMAT **Station chain race**	TIME **30–35 min**
GROUPING **Teams of 3–4**	TOPIC **Powers and Exponents (Flagship)**
STANDARD **6.EE.1**	PREP LEVEL **Medium — print puzzle strips and station cards**

Learning goal *I can evaluate powers to unlock the next station and create equivalent forms.*

Teacher setup

- Place numbered station cards around the room. Each answer is the next station number.
- Teams evaluate the power, then move to the station matching their answer.
- The final station asks students to create an equivalent repeated-multiplication card.

Materials

Puzzle strips, Station cards, Exponent check sheet, Pencil.

Differentiation & language support

- Check sheet lists base, exponent, and value columns (SPED).
- Sentence stem: “The value of ___ is ___, so we go to station ___.”
- ESOL: keep a worked power ($3^2 = 9$) visible as a model.

Key vocabulary (English / Español):

Term	Español	What it means
evaluate	evaluar	to find the value
equivalent	equivalente	equal in value

Quick teacher check: *Check $3^4 = 81$ on Card 3 — a common place for errors.*

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Name: **Date:** **Class:**

Your goal: *I can evaluate powers to unlock the next station and create equivalent forms.*

What you need

Puzzle strips, Station cards.

How to play

1. Start at Station 1 and evaluate the power.
2. The value is the number of the next station.
3. Continue until you finish the chain.
4. At the last station, create an equivalent repeated-multiplication card.

Station Powers

Evaluate each power. The value tells you which station to visit next.

1. Station 1: Evaluate 2^3.	2. Station 2: Evaluate 5^2.
3. Station 3: Evaluate 3^4.	4. Station 4: Evaluate 10^2.
5. Station 5: Evaluate 4^3.	6. Final: Write 2^5 as repeated multiplication and give its value.

Race log (power → value → next station):

Explain your thinking

Explain how the exponent tells you how many factors to multiply.

Sentence frame: *The value of ___ is ___ because the base is used ___ times.*

★ **Challenge:** *Create your own station clue: write a power, its value, and the next station number.*

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. Station 1: Evaluate 2^3. **Answer: 8 → go to Station 8**	**2.** Station 2: Evaluate 5^2. **Answer: 25 → go to Station 25**
3. Station 3: Evaluate 3^4. **Answer: 81 → go to Station 81**	**4.** Station 4: Evaluate 10^2. **Answer: 100 → go to Station 100**
5. Station 5: Evaluate 4^3. **Answer: 64 → go to Station 64**	**6.** Final: Write 2^5 as repeated multiplication and give its value. **Answer: 2 × 2 × 2 × 2 × 2 = 32**

Teaching notes & look-fors

- Values: $2^3 = 8$, $5^2 = 25$, $3^4 = 81$, $10^2 = 100$, $4^3 = 64$, $2^5 = 32$.
- $3^4 = 3 \times 3 \times 3 \times 3 = 81$ (not 12).